Tanvi Patil

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Data Workflow Engineer with 4+ years of experience building ML data pipelines, annotation quality frameworks, and automation tooling across AI-driven products. Currently at Apple supporting Vision Pro ML operations - designing end-to-end data workflows, Python automation, and dashboards that drive annotation accuracy and throughput. Published researcher in annotation quality prediction.

Education

San Diego State University — **San Diego, CA**
Master's degree in Information Systems, GPA: 3.9/4.0 — August 2022 - May 2024

Savitribai Phule Pune University — **Maharashtra, India**
Bachelor's degree in Computer Engineering, GPA: 3.6/4.0 — July 2017 - May 2021

Work Experience

Apple | Data workflow Engineer - Data Operations (Contract) — **September 2024 – Present**

- Managed end-to-end ML data workflows by validating/cleaning datasets and monitoring annotation pipelines, for high-priority projects.
- Built data submission pipelines, supported frame extraction workflows, and managed prelabel requests to improve data quality and efficiency.
- Developed Python automation scripts that reduced manual validation and submission effort by 40%, accelerating ML workflows.
- Wrote SQL queries and built dashboards to monitor data quality, throughput, and workflow bottlenecks for operational insights.
- Partnered with Operations, QA, and Engineering to optimize workflows and maintain 97% annotation accuracy across ML projects.
- Designed and deployed an internal project tracker to improve project visibility, task management, and cross-functional collaboration.

Think Digits | Data Analyst — **July 2023 – May 2024**

- Managed annotation projects for 1,000+ videos and images, producing consistent, high-quality datasets for AI/ML training.
- Designed data warehousing workflows (ETL, modeling, integration) to manage large datasets, improving accessibility and scalability.
- Implemented QC protocols that improved labeling accuracy to 95%, reducing rework and cutting model retraining costs.
- Collaborated with engineers to optimize data formats, enabling 30% faster ingestion into ML pipelines.

Amazon | Product Compliance Sr. Associate — **August 2021– July 2022**

- Conducted quantitative analysis in Python to validate compliance datasets, uncover patterns, and deliver insights for executive decision-making.
- Implemented inventory control processes to monitor 5,000+ SKUs, achieving 100% reporting accuracy for compliance audits.
- Built engaging dashboards and reports in Tableau/Excel that improved cross-functional communication by 40%, influencing product strategy.
- Applied SQL and Tableau to track compliance KPIs, driving continuous improvement initiatives that streamlined reporting workflows.

Typoline | Data Analyst — **January 2020 – January 2021**

- Streamlined data collection processes, cutting analysis time by 20% and boosting efficiency across reporting tasks.
- Applied statistical analysis and machine learning models (random forest, logistic regression) in Python/R to identify trends and support forecasting.
- Partnered with stakeholders to translate business needs into actionable, data-driven solutions.

Technical Projects

Transformative potential of large language models in healthcare

- Conducted thorough analysis of 20+ research papers and 10+ industry reports related to LLMs in healthcare.
- Investigated the use of prompt engineering techniques to enhance ethical considerations in LLM applications.
- Published a paper contributing significantly to the advancement of knowledge in healthcare

E-commerce Analyst: Predictive Modeling & Churn Classification (R)

- Conducted a comprehensive analysis on 100,000 orders from E-commerce, implementing feature engineering and predictive modeling to optimize e-commerce operations.
- Engineered key features, developed a powerful random forest model with 67.43% accuracy, identified customer churn patterns, and provided strategic recommendations for market growth.

Static Medical record using Blockchain.

- Proposed and developed a decentralized medical record storage application using Solidity, MetaMask, and blockchain technology.
- Implemented secure blockchain storage with smart contracts, ensuring data integrity and confidentiality.

Technical Skills & Tools

Programming Languages:	Python, R, HTML, JavaScript, React.js, C++
Libraries & Frameworks:	Pandas, NumPy, Selenium, PyTorch, SciPy, matplotlib, scikit-learn, seaborn.
Data Analysis & BI Software:	Tableau, Power BI, Advanced Microsoft Excel.
Databases:	MySQL, SQLite, NoSQL, MongoDB, Postgres.
Cloud & Technologies:	AWS (S3, EC2), SAS, GitHub.
Data Pipeline Tools:	DBT, ETL pipelines, Snowflake

Publications

- Paper Published, Retrieval-Augmented Generation and Fine-Tuning of Large Language Models in Healthcare: A Comparative Review with Applications (July 08, 2026). Available at SSRN: https://ssrn.com/abstract=7347718 or http://dx.doi.org/10.2139/ssrn.7347718
- Paper Published, Proactive Annotation Quality Prediction in Computer Vision Pipelines: A Metadata-Driven Framework (April 13, 2026). Available at SSRN: https://ssrn.com/abstract=6947180 or http://dx.doi.org/10.2139/ssrn.6947180
- Paper published, CareGraph: An Auditable Hybrid AI Framework for Evidence-Grounded Personalized Longitudinal Health Intelligence, Available here: https://arxiv.org/abs/2608.27484
- Patent published for the project 'Non-Hazardous Waste Sanitary Pad Decomposition System' on 3rd December 2021, Application No: 202021006596, Journal Number – 49/2021